MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours) Regular EXAMINATION October 2026, Semester V

Discipline Specific Elective (DSE) - MG5DSEBOTA00
APPLIED HORTICULTURE AND ENTREPRENEURSHIP

Duration: 2 Hours **Maximum Marks: 70**

Bloom's Taxonomy / Competency Keys: Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I), Appreciation (Ap)

PART A — SHORT ANSWER

Answer any 10 questions. Each question carries 2 marks.

1. What is meant by the method of manuring? [K / CO1]

The method of manuring refers to the application of organic matter to the soil to enhance its nutrient content, improve structure, and support healthy plant growth.

2. Comment on the commercial application of gibberellins. [K / CO1]

Seed germination, malt production, grape sizing.

3. What is an imitative entrepreneur? State one of its characteristics. [K / CO3]

Definition: A person who adopts or copies successful business ideas, products, or methods developed by other entrepreneurs. (1 mark)
Characteristics: Copies successful business ideas from others, adopts existing technologies and methods, takes less risk than an innovative entrepreneur. (Any 1 point - 1 mark)

4. Name one tool used for flower arrangement. [K / CO4]

Floral scissors, pruning shears, etc. (Any one tool used for flower arrangement - 2 marks).

5. Mention any two value-added products obtained from jackfruit. [U / CO3]

Jackfruit chips and jackfruit leather (or chakkavarattayith) or any such items

6. What is Integrated Weed Management? [K / CO2]

Planned use of two or more compatible weed-control methods to keep weeds below the level at which they cause significant economic loss. It includes cultural, mechanical, physical, chemical, and biological methods.

7. Define leadership. [K / CO3]

Ability of an entrepreneur to guide, motivate, and influence employees to work together towards achieving business goals.

8. How farm tourism helps farmers? [K / CO4]

Creates additional income for farmers, rural employment, direct sale of farm products, employment opportunities, better use of farm resources, women's empowerment. (Any 2 valuable points)

9. What are the main components of soil? [K / CO1]

The main components of soil are mineral particles, organic matter, water, and air. microorganisms and humus

10. Name two plant growth regulators used to promote rooting in cuttings. [U / CO2]

Auxin, cytokinin, IAA etc

11. Explain how KIED supports small and aspiring entrepreneurs. [U / CO3]

Kerala Institute for Entrepreneurship Development (KIED) supports small and aspiring entrepreneurs by conducting structured entrepreneurship development programs and providing incubation, mentoring, and financial facilitation services.

12. Name one mushroom species commonly used for cultivation. [K / CO4]

Oyster mushroom (Pleurotus ostreatus) (Any one edible mushroom - 2 marks).

[2 x 10 = 20 Marks]

PART B — SHORT ESSAYS

Answer any 8 questions. Each question carries 5 marks.

13. Write a short note on leaf mould and its importance in improving soil fertility. [U / CO1]

- Source: Decayed forest tree leaves and leaf litter.
- Properties: Highly porous, soft, acid-neutral conditioner.
- Usage: Ideal for nurseries, potted crops, and horticulture
- Leaf mould is a natural soil conditioner produced by the slow decomposition of fallen autumn leaves, primarily through fungal action

14. Explain the economic importance of commercial floriculture in India. [K / CO2]

1. Generates higher returns per unit area.
2. Creates employment (nursery production, packing, transportation, etc.).
3. Industrial importance: Commercial flowers serve as raw materials for perfumes, cosmetics, natural dyes, pharmaceutical, and herbal industries.
4. Importance of cut and loose flowers.
5. Cultural importance of flowers in India for festivals and other occasions. (Any 5 valuable points - 5 marks)

15. Differentiate between novice, serial, and portfolio entrepreneurs. [K / CO3]

- Novice entrepreneurs: A person who starts a business for the first time with little or no previous business ownership experience.

- Serial entrepreneurs: A person who starts one business after another, gaining experience through starting and developing different businesses.
- Portfolio entrepreneurs: A person who owns and manages several businesses at the same time, gaining experience by managing multiple businesses.

16. Explain the steps of making herbal cosmetic products. [U / CO4]

1. Selection of herbs: Choose suitable medicinal plant.

2. Collection and cleaning: Collect fresh healthy plant parts and wash thoroughly.
3. Processing of ingredients: Dry, grind, crush, or extract the useful components.

4.Formulation: Mix the herbal extracts with suitable ingredients.
5.Quality testing: Check product appearance, pH, stability, and microbial safety.
6. Packaging: Pack finished products in clean, suitable containers to protect from contamination.
7. Labeling and storage: Label with ingredients, manufacture date, expiry/best-before info, and storage instructions. Store under suitable conditions.

17. Give a definition of olericulture and mention its major areas of application. [K / CO1]

It is the production utilization and improvement of vegetables, which produces nutrient-rich vegetables for human consumption.

- Ensures food and nutritional security.
- Rich source of vitamins, minerals, dietary fibre, and antioxidants.
- Generates high income and employment opportunities.
- Supplies raw materials for food processing industries.
- Contributes to export earnings and rural development.

18. Discuss the grading, packing, and storage practices for gerbera cut flowers. [K / CO2]

- Grading: Based on stalk length, stalk thickness and strength, flower diameter, freedom from pests, diseases, mechanical damage, flower form, and colour.
- Packing: Flower stalks are soaked in solutions such as sodium hypochlorite; flowers are sorted into uniform lots, aligned, and bundled. Packaging should prevent bending, crushing, moisture loss, and mechanical injury.
- Storage: Pre-cooled soon after harvest, stored at low temperature and high relative humidity. Clean water or holding solutions maintain hydration. (Any 5 valuable points - 5 marks)

19. Explain the role of DPIIT in developing and supporting the Indian startup ecosystem. [U / CO3]

The Department for Promotion of Industry and Internal Trade (DPIIT), under the Ministry of Commerce and Industry, serves as the nodal agency driving the flagship Startup India initiative. It grants official DPIIT recognition to eligible startups, unlocking vital benefits such as income tax exemptions, fast-tracked patent applications at reduced costs, and self-certification under basic labor and environmental laws.

20. Explain the spatial zoning of a kitchen garden. [K / CO1]

- Zone 1 (Immediate Kitchen Access): Reserve the area within 5–10 steps of the kitchen door for high-frequency harvest items—Curry Leaf, *Kanthari* chili, *Panikoorkka*, Mint, and coriander/microgreens.
- Zone 2 (Sunlit Garden Beds): Position raised beds or grow bags in spots receiving 6+ hours of sunlight for sun-loving seasonal vegetables like tomatoes, okra (*Vendakka*), and gourds.
- Zone 3 (Perimeter & Canopy): Place tall fruit trees and coconut palms along property boundaries to serve as windbreaks and privacy screens without overshadowing the core vegetable zones.

21. What is a net house? Describe its structure and uses in plant propagation. [U / CO2]

- Definition: A protected cultivation structure covered with shade net, insect-proof net, or synthetic mesh materials to protect plants from sunlight, insects, wind, and environmental stresses while allowing air circulation. (2 marks)
- Structure: Supporting framework (Galvanized iron, steel, bamboo), Net covering (shade/insect-proof nets over roof and sides), entrance, and growing area. (1 mark)
- Uses: Raising nursery seedlings, production of shade-loving ornamental plants, protection from insect pests/birds, hardening tissue-cultured plants, producing healthy planting material. (2 marks)

22. Explain the major activities involved in implementing a business plan. [U / CO3]

1. Arranging Finance
2. Organizing Resources
3. Recruiting Employees
4. Starting Business Operations
5. Monitoring and Evaluation (Short description of any 5 valuable points - 5 marks)

[5 x 8 = 40 Marks]

PART C — ESSAY

Answer any 1 question. Each question carries 10 marks.

23. Discuss the fundamental principles involved in gardening. [K / CO1]

There are certain basic gardening principles that a gardener should follow to achieve a garden pleasing to the senses and provides a pleasant outdoor living space. They include, **Balance., Proportion, Unity, Perspective, Prospect, Restraint, Rhythm, Harmony, Surprise, Scale, Space** etc

1. The balance in landscape design is visual equilibrium of different garden elements.
2. Proportion refers to the share of the different parts or components to the whole. It is the relation of the component with other in magnitude
3. Unity means, landscape should achieve unity by ensuring that all plants, structures, and landscape features complement one another and the main building

4. Perspective is the visual effect in which objects appear smaller as their distance from the viewer increases (railway tracks, which appear to converge at the horizon despite remaining parallel) and this phenomenon of apparent size reduction and line convergence
5. Prospect is the view of a scenery, natural or man made, through an opening such as window or a gap in the foliage of trees.
6. Restraint: Overuse of any component including grouping of plants in a particular location masks the scenic beauty.
7. Rhythm is measured as cyclic repetition of an object, effect and event.
8. Harmony is the pleasing effect created by the appropriate arrangement and combination of various garden features.
9. Surprise is an important landscape design principle that maintains curiosity and interest
10. Scale is a relative dimension. The height and spread of trees and shrubs and the spread of the water garden are determined by adopting a scale.

(Give full marks to short explanations of any 8 features)

24. Describe the concept of training in crop plants. Discuss its objectives and various methods with suitable examples. [K / CO2]

Definition: Training is the practice of directing and controlling the growth of a young fruit plant into a desired shape or framework by tying, supporting, bending, or selectively removing shoots. (2 marks)

Objectives of Training: (Any 2 points - 2 marks)

- Develop a strong framework capable of bearing heavy fruit loads.
- Maintain proper tree shape and size.
- Improve light penetration and air circulation.
- Facilitate orchard operations (pruning, spraying, harvesting, intercultural operations).
- Increase photosynthetic efficiency and reduce wind/load breakage.
- Improve fruit colour, size, quality, and pest management.

Methods of Training: (Any 4 methods with examples - 6 marks)

Open Centre System (Vase System): Central leader removed after planting; 3-5 scaffold branches grow outward forming an open centre for better sunlight. Eg: Apple, peach, mango, guava.

Central Leader System: Main trunk grows uninterrupted; lateral branches arise regularly, forming a pyramidal shape with strong structural support. Eg: Pear, walnut, sapota.

Cordon System: Single strong shoot grown to desired height, then bent and tied horizontally along supporting wires to form permanent arms maintained by annual pruning. Eg: Grapes, passion fruit, vanilla.

Head System (Bush System): Main trunk kept short (30-90 cm). 4-6 primary branches arise from the head giving a bush-like appearance. Eg: Grapes, pepper.

Bower (Pergola) System: Vines grow over an overhead network of wires supported by pillars to form a roof-like canopy. Eg: Grapes, passion fruit, bitter gourd.

[10 x 1 = 10 Marks]